



BUSINESS REQUIREMENT DOCUMENT

Executive Summary

1. Engagement & Purpose: A comprehensive operational audit and software feasibility study was conducted for Manugiants Pvt. Ltd. to design a custom digital inventory management application aimed at digitizing the organization's manual supply chain workflows.

2. Core Problems: The audit identified two critical failure points: severe data latency caused by a 24-hour manual ledger update cycle and high operational risk due to a significant discrepancy between physical stock counts and digital records.

3. Definition of Success: Success will be defined by the successful deployment of the Manugiants Warehouse Management App, achieving a 99% inventory accuracy rate and a 40% reduction in order fulfillment time within the first two quarters of implementation.

Business Objectives

BO-01: Digital Transformation of Inventory Tracking Eliminate 100% of paper-based ledgers by transitioning all stock-in and stock-out activities to a digital interface, ensuring data integrity across all departments.

BO-02: Real-Time Data Synchronization Reduce data latency from 24 hours to near-instantaneous (under 30 seconds) to provide the Sales team with accurate "Available-to-Promise" (ATP) stock levels.

BO-03: Reduction in Order Processing Time Streamline the "Pick & Pack" workflow to reduce the total time from order receipt to dispatch by 40%, directly addressing current shipping delays and client dissatisfaction.



BO-04: Operational Cost Optimization Decrease labor costs associated with manual reconciliations and physical stock checks by automating balance updates, allowing staff to focus on high-value logistics tasks.

Current State Problem

The Manual Inventory Tracking Crisis

The current inventory management framework at Manugiants Pvt. Ltd. relies entirely on a decentralized, paper-based ledger system. Every stock movement-whether receiving new shipments or picking items for orders-is recorded manually on paper slips before being transcribed into a master Excel sheet at the end of the business day. This workflow creates a 24-hour data lag, meaning the "available stock" visible to the sales team is consistently inaccurate. This lack of a "Single Source of Truth" leads to frequent "ghost inventory" errors, where items are sold digitally that do not exist physically on the warehouse shelves.

Procurement and Restock Inefficiency

Because there is no real-time tracking or automated low-stock alerts, the procurement process is purely reactive rather than proactive. Stock-outs are often only discovered during the physical "Pick & Pack" phase, long after a customer has placed an order. This manual "shelf-check" dependency prevents the procurement team from identifying trends or maintaining optimal reorder points. Consequently, the warehouse fluctuates between extreme overstocking of slow-moving goods and critical shortages of high-demand items, tying up capital and stalling the supply chain.

B2B Client Attrition and Shipping Delays

The cumulative effect of these systemic failures is a direct hit to the company's reputation and bottom line. B2B clients, who rely on



predictable delivery schedules to maintain their own operations, are frequently facing unannounced shipping delays of 48 hours or more due to inventory discrepancies. This unreliability has moved beyond a minor inconvenience to a major driver of customer attrition. Without a digital system to provide "Available-to-Promise" (ATP) accuracy, Manugiants is losing high-value contracts to competitors who can guarantee real-time stock confirmation and faster fulfillment cycles.

SCOPE

In-Scope (The "Must-Haves")	Out-of-Scope (Excluded)	Reason for Exclusion
Development of a cross-platform (iOS/Android) Warehouse App.	Integration with External Courier APIs.	Prioritizing internal warehouse efficiency before external tracking.
Barcode/QR scanning functionality for stock movement.	Customer-facing Order Tracking Portal.	Current phase focuses on solving "Ghost Inventory" for staff only.
Real-time inventory dashboard for Sales/Admin teams.	Marketing and Lead Generation tools.	These do not solve the manual ledger discrepancy issue.
Automated low-stock alerts and procurement triggers.	Historical Data Migration (pre-2024).	To ensure system speed; only current active inventory will be migrated.



Stakeholder List

Stakeholder	Role	Influence (1-10)
Operations Head	Project Sponsor	10
Warehouse Staff	Primary Users	4
Procurement Manager	Functional Lead	7
IT Manager	Technical Lead	8
Logistics Manager	Process Owner	6

Note: For detailed contact information and communication preferences, refer to **Appendix A: Full Stakeholder Matrix**.



Business Requirements

ID	Business Requirement Description	Priority	Stakeholder Owner
BR-01	The system shall update inventory levels in real-time upon barcode/QR scan during picking and packing.	Critical	Warehouse Manager
BR-02	The system shall trigger an automatic reorder alert to the Procurement Lead when stock falls below a defined safety threshold.	High	Procurement Lead
BR-03	The system shall generate a daily fulfillment dashboard highlighting pending, shipped, and delayed orders for the Logistics Manager.	High	Logistics Manager
BR-04	The system shall require secure user authentication (Login/Password) and provide Role-Based Access Control (RBAC).	Critical	IT / Admin
BR-05	The system shall allow the Sales team to view "Live Stock"	Critical	Sales Team



	availability during the order entry process to prevent over-selling.		
BR-06	The system shall capture a digital "audit trail" (timestamp and user ID) for every manual stock adjustment.	Medium	Operations Head
BR-07	The system shall generate weekly "Stock Turnover Reports" to identify slow-moving vs. fast-moving goods.	Medium	Procurement Lead
BR-08	The system shall provide an automated "Goods Receipt" module to digitize incoming shipments from suppliers instantly.	High	Warehouse Staff



FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Functional Requirements

FR ID	Description	Linked BR
FR-01	The system shall update inventory levels in real-time (within 3 seconds) upon a successful barcode/QR scan during picking and packing operations.	BR-01
FR-02	The system shall automatically trigger a reorder alert notification to the Procurement Lead within 60 seconds when any SKU falls below its defined safety stock threshold.	BR-02
FR-03	The system shall generate and display a daily fulfillment dashboard categorizing all orders as Pending, Shipped, or Delayed, refreshed every 24 hours.	BR-03
FR-04	The system shall authenticate all users via username/password and enforce Role-Based Access Control (RBAC), restricting module access based on assigned user role.	BR-04



FR-05	The system shall display real-time available stock quantity to Sales Team members at the point of order entry to prevent over-selling.	BR-05
FR-06	The system shall automatically generate a digital audit trail — capturing User ID, timestamp, and quantity changed — for every manual stock adjustment.	BR-06
FR-07	The system shall generate a weekly Stock Turnover Report ranking all SKUs by movement rate, classifying items as fast-moving or slow-moving.	BR-07
FR-08	The system shall provide a Goods Receipt module that digitally records incoming supplier shipments via barcode scan and immediately updates inventory.	BR-08



Non-Functional Requirements

NFR ID	Category	Description
NFR-01	Performance	Inventory levels must update within 3 seconds of a confirmed barcode scan under normal network conditions.
NFR-02	Availability	The system must maintain 99.5% uptime during core warehouse operating hours (6 AM – 10 PM IST).
NFR-03	Security	All data must be encrypted at rest (AES-256) and in transit (TLS 1.2+). Admin accounts must enforce Multi-Factor Authentication (MFA).
NFR-04	Scalability	The system must support a minimum of 25 concurrent users without performance degradation.
NFR-05	Usability	Warehouse Staff must be able to complete core scanning workflows after no more than 2 hours of training.
NFR-06	Compatibility	The application must function on industrial-grade Android handheld scanners and standard web browsers (Chrome, Edge — latest 2 versions).
NFR-07	Maintainability	The system codebase must follow modular architecture to allow feature updates without full redeployment.
NFR-08	Data Integrity	All migrated stock records must pass automated validation checks; a 20% manual spot-check against original paper ledgers is mandatory pre-Go-Live.



Assumptions & Constraints

Category	Description
Assumption	It is assumed that the warehouse management team has validated the current physical stock count prior to the digital migration cutover.
Dependency	Hardware Procurement: The project requires 15 industrial-grade handheld scanners. Note: Procurement is managed under a separate Capital Expenditure (CAPEX) budget of ₹1,50,000, independent of the software development budget.
Dependency	Pre-Validation of Labels: Prior to development, a sample of supplier barcode/QR formats must be validated against the chosen scanning library to ensure compatibility.
Constraint	Budget: The development budget is capped at ₹4,50,000, covering software engineering, cloud hosting, and licensing only.
Constraint	Timeline: The Minimum Viable Product (MVP)-focusing on core scanning and real-time updates-must be deployed within a 90-day window.



RISKS

Risk ID	Risk Description	Probability (H/M/L)	Impact (H/M/L)	Risk Owner	Mitigation Strategy
R-01	Staff resistance to digital transition.	High	High	Warehouse Manager	Appoint "Change Champions" from each shift; run a 1-week pilot program for feedback before full rollout.
R-02	Wi-Fi "dead zones" preventing real-time sync.	Medium	High	IT Lead	Conduct a site-wide signal audit; install mesh routers as per the project dependency.
R-03	Data Corruption during Migration: Errors during the transfer of paper records to digital format.	Medium	High	Data Analyst	Implement a 100% automated validation script for digital entries; perform a 20% manual spot-check against original paper ledgers.
R-04	Incompatible Label Formats: Supplier barcodes fail to scan with standard libraries.	Medium	High	Lead Developer	Execute a "Scanner Pilot" during the design phase using physical supplier samples; develop custom parsing logic as a contingency.
R-05	Hardware Procurement Delay: Scanners are not delivered before the 90-day	Low	High	Project Manager	Confirm vendor lead times immediately. Contingency: Temporary use of high-end smartphones for low-



	software deployment.				volume SKUs only; full warehouse operation is impossible without industrial scanners.
R-06	Unauthorized adjustments: Security breach of inventory levels.	Low	High	IT Lead	Implement Role-Based Access Control (RBAC) and Multi-Factor Authentication (MFA) for all administrative accounts.
R-07	System Integration Failure: API or data-flow failure between the App and existing ERP/Accounting software.	High	High	IT Lead	Conduct 4 weeks of sandbox environment testing with dummy data; establish a "Manual Sync" protocol as a temporary backup during the first week of Go-Live.

Sign-Off

Name	Role	Signature	Date
Operations Head	Project Sponsor		
IT Manager	Approver		